Status	Finished
Started	Thursday, 8 October 2026, 7:45 PM
Completed	Thursday, 8 October 2026, 7:57 PM
Duration	12 mins 26 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It can only be applied to large lists
c. It does not work with negative numbers
d. It requires the list to be sorted ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the list contains duplicate elements
c. When the list is sorted
d. When the search algorithm finishes

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Merge search
c. Linear search ✓
d. Quick search

Question **4**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n ✓
- b. log n
- c. 1
- d. n/2

Question **5**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and sorted
- b. When the list is small and unsorted ✓
- c. When the list is large and unsorted
- d. When the list is large and sorted

Question **6**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The list is sorted
- b. The search ends successfully ✓
- c. The search continues in the left sublist
- d. The search continues in the right sublist

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The last element is returned
- b. The first element is returned
- c. The search is considered unsuccessful ✓
- d. An error is raised

Question **8**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Sorting elements in a list
- c. Determining whether an element is present in a list ✓
- d. Inserting elements into a list

Question **9**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O(log n)
- c. O👎
- d. O(n log n)

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(n log n)
- c. O(log n) ✓
- d. O(1)

Question **11**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the first element in the list
- b. Compare the target element with the middle element in the list ✓
- c. Sort the list
- d. Divide the list into two equal parts

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By sorting the list first
c. By using a hash function
d. By comparing each element in the list sequentially ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search does not require dividing the list
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search can find multiple instances of the target element

Question **14**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Complexity

c. Bubble ✓

d. Insertion

Question **15**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering

b. Mining

c. Finding

d. Searching ✓